

	ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		Assignment
Branch	CSE/CSIT	Programme	B.Tech
Course Name	Compilers: Principles, Techniques and Tools	Semester	6th
Course Code	CSE 3739	Academic Year	2026
ASSIGNMENT - I			
Submission due date: 02/04/2026			
Learning Level (LL)	L1: Remembering	L3: Applying	L5: Evaluating
	L2: Understanding	L4: Analysing	L6: Creating
Q's	Questions		CO1
1	What are the advantages of an interpreter over a compiler?		CO1
2	Consider the following expression: $id1 = id2 + id3 * 10$. Describe the input and output of all phases of a compiler.		CO1
3	Why is Java called a Compiler-Interpreter Language?		CO1
4	Find out all the tokens for the following 'C' program. <pre> unsigned int reverseBits(unsigned int n) { unsigned int reversed = 0; int numBits = sizeof(n) * CHAR_BIT; for (int i = 0; i < numBits; i++) { if ((n & 1) != 0) { reversed = 1, U << (numBits - 1 - i); } // Right shift n by 1 to process the next bit n >>= 1; } return reversed; } </pre>		CO2
5	Draw the transition diagram for a valid hexadecimal constant in the C programming language.		CO2
6	Construct the regular definition for the above transition diagram (Considering question no. 5).		CO2
7	Consider the Context-free Grammar: $E \rightarrow E + E \mid E * E \mid E \uparrow E \mid id$ Construct an unambiguous grammar from the above CFG.		CO2
8	Consider the following Context-free Grammar G: $A \rightarrow abd \mid abde \mid abdA \mid abdAe \mid \epsilon$ Remove the left factor and write the modified grammar.		CO3
9	Let G be the following Context-free Grammar, $S \rightarrow ASa \mid aBa \mid bA \mid C$ $A \rightarrow aB \mid dS$ $B \rightarrow CD \mid aBd$ $C \rightarrow cDd \mid \epsilon$ $D \rightarrow d$ Compute the FIRST and FOLLOW functions for all the non-terminals present in the above Grammar. Check the above		CO3

	grammar for LL(1) by using the predictive parsing table.		
10	Let G be the following Context-free Grammar $A \rightarrow (A) \mid a$ Parse the string “(((a)))” using a shift-reducing parsing technique.	CO3	L3,L5

Course Outcomes	By the end of the course, through lectures, readings, home works, assignments, and exams, students will be able to:	
	CO1	Summarize the overview of programming languages, language processors and the structure of a compiler
	CO2	Acquire knowledge in theory of computation and their role in designing different types of tokens generated by lexical analyzer
	CO3	Demonstrate the role of Parser(s) (LL, SLR, CLR and LALR) and its types i.e. Top-down and Bottom-up parsers.
	CO4	Apply and evaluate syntax directed translation schemes, synthesized attributes, inherited attributes, and different techniques for symbol table organization
	CO5	Analyze the generation of various intermediate codes and the process of their optimization.
	CO6	Examine the target machine's run time environment, and its instruction set for code generation and techniques used for code optimization

- ✓ Assignment scores/markings depend on neatness and clarity.
- ✓ Plagiarized assignments will be given a zero mark.
- ✓ Submit the hard copy of your assignment by the due date.
- ✓ Submit the assignment handwritten on A4 size papers and soft tape bound to your CPTT class teacher. A front page must be present containing the details of the subject, the assignment and the student.